

RAJHANSH YADAV

AI/ML Developer | Python | Machine Learning | BCA (AI/ML) Student

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SUMMARY

BCA (AI/ML) student at Noida International University who learns by building projects from first principles rather than only following tutorials. Built a neural network from scratch in NumPy (98.1% test accuracy on handwritten digits, backpropagation verified with a numerical gradient check), an end-to-end ML pipeline served through a FastAPI REST API and deployed with Docker and CI, and a Python execution tracer that estimates time complexity automatically. Strong foundation in data structures & algorithms, object-oriented programming and core CS; looking for AI/ML internship opportunities.

SKILLS

- Languages: Python, C, Java
- AI / ML & Data: NumPy, PyTorch, Hugging Face, scikit-learn, Matplotlib, FastAPI
- Core CS: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Architecture
- Web & Tooling: Next.js, React, Tailwind CSS, REST APIs, Vercel
- Dev Tools: Git/GitHub, GitHub Actions (CI), Docker, pytest, Google Colab, VS Code

PROJECTS

NeuralFlow-AI — End-to-End ML Pipeline for Breast-Tumor Classification [GitHub →](#)

- Built a complete pipeline comparing Logistic Regression, Random Forest and a Neural Network on the Breast Cancer Wisconsin (Diagnostic) dataset, selecting the best model by cross-validated metrics and tuning the decision threshold for the precision/recall trade-off.
- Served the trained model through a FastAPI REST endpoint, containerised the app with Docker, and wrote a pytest suite with GitHub Actions CI so every push is tested automatically.

Tech: Python, scikit-learn, FastAPI, Docker, pytest, GitHub Actions

Neural Network From Scratch (NumPy) [GitHub →](#)

- Built a multi-layer perceptron (64→64→32→10) using only NumPy — forward pass, backpropagation, Adam optimiser and L2 regularisation — to classify handwritten digits. Reached 98.1% test accuracy vs. 96.7% for a logistic-regression baseline.
- Verified backpropagation with a numerical gradient check (worst relative error 4.0e-07); avoided data leakage by scaling with train-only statistics and a stratified train/validation/test split. Added pytest tests and GitHub Actions CI, plus training curves, a confusion matrix and error analysis.

Tech: Python, NumPy, scikit-learn, Matplotlib, pytest, GitHub Actions

Scope Visualizer — Python Execution Tracer & Complexity Estimator [GitHub →](#)

- Built a tool that records every function call (arguments, depth, time, memory) using sys.setprofile and tracemalloc, and writes one self-contained HTML report with a timeline, memory curve and collapsible call tree.
- Added an empirical complexity estimator that times a function at growing input sizes and fits $O(1)$ to $O(n^3)$ models; correctly identified $O(n)$, $O(n \log n)$ and $O(n^2)$ for linear sum, merge sort and bubble sort. Tested with pytest and GitHub Actions CI.

Tech: Python, sys.setprofile, tracemalloc, HTML/SVG, pytest, GitHub Actions

journey-to-job-ai-ml — Public Learning Roadmap [GitHub →](#)

- Maintain a public, commit-by-commit roadmap tracking Python internals, data structures and production-grade ML architecture as I learn them.

Tech: Python 3.12, PyTorch, Hugging Face, Git/GitHub

EXPERIENCE

Freelance Developer — Local food businesses

- Built WhatsApp automation bots for local food businesses to automate customer communication and support business growth.
- Designed and developed a website for a local pizza shop.

EDUCATION

Bachelor of Computer Applications (AI/ML Specialisation) 2025 – 2028

Noida International University, Uttar Pradesh — currently in 3rd semester

CERTIFICATIONS

- Generative AI Essentials & AI and Cybersecurity Awareness — TCS iON, “AI for All” initiative by MPIT–CoE & TCS Foundation (2026)
- Python 101 for Data Science (PY0101EN) — IBM / Cognitive Class (2026)
- Python — Kaggle (2026)